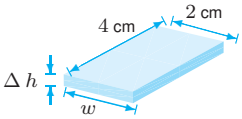


(b) $8(1 - h/3)\Delta h$;
$\int_0^3 8(1 - h/3)\,dh = 12\ \text{cm}^3$



37 $\int_0^{150} 1400(160 - h)\,dh = 1.785 \cdot 10^7\ \text{m}^3$

39 Change to $\int_{-10}^{10} \pi\left(10^2 - x^2\right)\,dx$

41 Region between positive x-axis, positive y-axis, and $y = 1 - x$

43 False

45 True

Section 8.2

1 (a) $4\pi \int_0^3 x^2\,dx$
(b) 36π

3 (a) $(\pi/4)\int_0^6 (36 - y^2)\,dy$
(b) 36π

5 $\pi/5$

7 $256\pi/15$

9 $\pi(e^2 - e^{-2})/2$

11 $\pi/2$

13 $2\pi/15$

15 2.958

17 2.302

19 π

21 $\sqrt{42}$

23 ≈ 24.6

25 $V = \int_0^2 [\pi(9 - y^3)^2 - \pi(9 - 4y)^2]\,dy$

27 $V = \int_0^9 [\pi(2 + \frac{1}{3}x)^2 - \pi 2^2]\,dx$

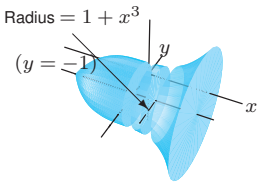
29 $V = \int_0^5 \pi((5x)^2 - (x^2)^2)\,dx$

31 $V = \int_0^5 \pi((4 + 5x)^2 - (4 + x^2)^2)\,dx$

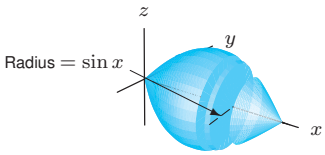
33 3.820

35 (a) $16\pi/3$
(b) 1.48

37 $V = (16/7)\pi \approx 7.18$



39 $V = (\pi^2/2) \approx 4.935$



41 $4\pi/5$

43 $8/15$

45 $\sqrt{3}/8$

47 $V \approx 42.42$

49 $V = (e^2 - 1)/2$

51 $g(x) = 2\sin x, -2\sin x$

53 2267.32 cubic feet

55 (a) Volume $\approx 152\ \text{in}^3$
(b) About 15 apples

57 $40,000LH^{3/2}/(3\sqrt{a})$

59 (a) $dh/dt = -6/\pi$
(b) $t = \pi/6$

61 (a) $4\int_0^r \sqrt{1 + (-x/y)^2}\,dx$
(b) $2\pi r$

63 $\int_0^4 \sqrt{1 + (4 - 2x)^2}\,dx$

65 $\int_{-\sqrt{1/2}}^{\sqrt{1/2}} \sqrt{1 + 4x^2 e^{-2x^2}}\,dx$

67 4.624 meters

69 (c) $f(x) = \sqrt{3}x$

71 Change to $\int_0^5 \left(\pi(3x)^2 - \pi(2x)^2\right)\,dx$

73 Greater than 32

75 Region bounded by $y = 2x$, x-axis, $0 \le x \le 1$

77 $f(x) = x^2$

79 False

81 False

Section 8.3

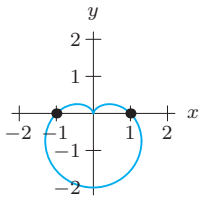
1 $(-1/2, \sqrt{3}/2)$

3 $(3, -\sqrt{3})$

5 $(\sqrt{2}, \pi/4)$

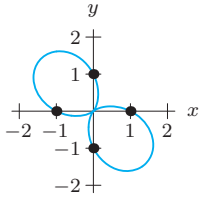
7 $(2\sqrt{2}, -\pi/6)$

9 (b)



(c) Cartesian:
$(\sqrt{3}/4, 1/4)$;
$(-\sqrt{3}/4, 1/4)$ or polar:
$r = 1/2$, $\theta = \pi/6$ or $5\pi/6$

(d)



11 Looks the same

13 Rotated by 90° clockwise

15 $\pi/4 \le \theta \le 5\pi/4$;
$0 \le \theta \le \pi/4$ and $5\pi/4 \le \theta \le 2\pi$

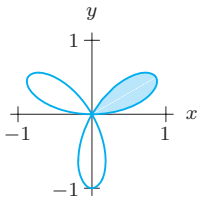
17 $\sqrt{8} \le r \le \sqrt{18}$ and $\pi/4 \le \theta \le \pi/2$

19 $0 \le \theta \le \pi/2$ and $1 \le r \le 2/\cos\theta$

21 -1

23 $\sqrt{2}(e^{\pi} - e^{\pi/2})$

25



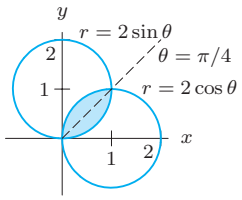
27 $4\pi^3$

29 (a) $r = 1/\cos\theta$; $r = 2$
(b) $\frac{1}{2}\int_{-\pi/3}^{\pi/3} (2^2 - (1/\cos\theta)^2)\,d\theta$
(c) $(4\pi/3) - \sqrt{3}$

31 $2\pi a$

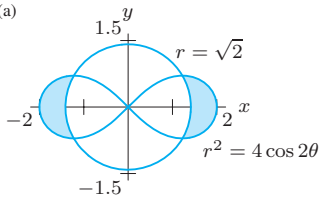
33 $(5\pi/6) + 7\sqrt{3}/8$

35 (a)



(b) $(\pi/2) - 1$

37 (a)



(b) $2\sqrt{3} - 2\pi/3$

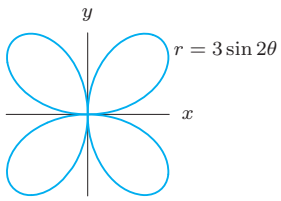
39 Horiz: $(\pm 1.633, \pm 2.309)$; $(0, 0)$
Vert: $(\pm 2.309, \pm 1.633)$; $(0, 0)$

41 (a) $y = (2/\pi)x + (2/\pi)$
(b) $y = 1$

43 21.256

47 2.828

49 Points are in quadrant IV

51 

53 $r = 100$

55 $r = 1 + \cos\theta$

Section 8.4

1 $1 - e^{-10}$ gm

3 (a) $\displaystyle\sum_{i=1}^{N}(2 + 6x_i)\Delta x$

(b) 16 grams

5 (b) $\sum_{i=1}^{N}[600 + 300\sin(4\sqrt{x_i + 0.15})](20/N)$

(c) ≈ 11513

7 2 cm to right of origin

9 45

11 186,925

13 1 gm

15 (a) 3 miles

(b) 282,743

17 (a) $\int_0^5 2\pi r(0.115e^{-2r})dr$

(b) 181 cubic meters

19 (a) $\pi r^2 l/2$

(b) $2klr^3/3$

21 $\int_0^{60}(1/144)g(t)\,dt$ ft^3

23 $x = 2$

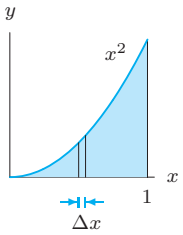
25 $\pi/2$

27 (a) Right

(b) $2/(1 + 6e - e^2) \approx 0.2$

29 (a) 2/3 gm

(b) Greater than 1/2



(c) $\overline{x} = 3/4$ cm

33 $\overline{x} = (1 - 3e^{-2})/(2 - 2e^{-2}) \approx 0.343$; $\overline{y} = 0$

35 Density cannot be negative

37 $\delta(x) = x^2$, for $-1 \le x \le 1$

39 $\delta(x) = \begin{cases} 5, & 0 \le x \le 1 \\ 1, & 1 < x < 5 \\ 4, & 5 \le x \le 10 \end{cases}$

41 $\delta(x) = 1$

43 False

45 True

47 True

Section 8.5

1 30 ft-lb

3 1.333 ft-lb

5 27/2 joules

7 $1.176 \cdot 10^7$ lb

9 20 ft-lb

11 $1.489 \cdot 10^{10}$ joules

13 3437.5 ft-lbs

15 6860 joules

17 784.14 ft-lb

19 1,058,591.1 ft-lbs

21 (a) 156,828 ft-lbs

(b) 313,656 ft-lbs

(c) 150,555 ft-lbs

23 27,788 ft-lbs

25 $0.366(k + 1.077)g\pi$ joules

27 (a) Force on dam $\approx \sum_{i=0}^{N-1} 1000(62.4h_i)\Delta h$

(b) $\int_0^{50} = 1000(62.4h)\,dh =$ 78,000,000 pounds

29 4,992,000 lbs

31 (a) $1.76 \cdot 10^6$ nt/m^2

(b) $1.96 \cdot 10^7 \int_0^{180} h\,dh = 3.2 \cdot 10^{11}$ nt

33 (a) 780,000 lb/ft^2
About 5400 lb/in^2

(b) $124.8\int_{-3}^{3}(12{,}500 - h)\sqrt{9 - h^2}\,dh$
$= 2.2 \cdot 10^7$ lb

35 Potential $= 2\pi\sigma(\sqrt{R^2 + a^2} - R)$

37 15231 ergs

39 $\dfrac{GM_1M_2}{l_1l_2}\ln\left(\dfrac{(a + l_1)(a + l_2)}{a(a + l_1 + l_2)}\right)$

41 $(2GMmy/a^2)\left(y^{-1} - (a^2 + y^2)^{-1/2}\right)$

43 Less than twice as much

45 Book raised given vertical distance

47 False

49 False

51 False

Section 8.6

1 $C(1 + 0.03)^{25}$ dollars

3 $C/e^{0.03(5)}$ dollars

5 $\int_0^{15} C\,e^{0.02(15-t)}\,dt$ dollars

7 $\ln(25{,}000/C)/30$ per year

9 (a) Future value = \$6389.06
Present value = \$864.66

(b) 17.92 years

11 (a) \$43, 645.71

(b) \$20, 000.00

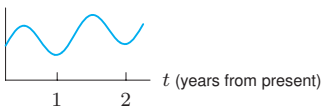
(c) \$23, 645.71

13 4.621%/ yr

15 \$4000/ yr; \$21,034.18

17 \$1000/ yr; \$24,591.23

19 \$/year



21 (a) \$5820 per year

(b) \$36,787.94

23 Installments

25 3.641%/ yr

27 9.519%/ yr

29 (a) Option 1

(b) Option 1: \$10.929 million;
Option 2: \$10.530 million

31 In 10 years

33 (a) $\sum_{i=0}^{n-1}(2000 - 100t_i)e^{-0.1t_i}\Delta t$

(b) $\int_0^M e^{-0.10t}(2000 - 100t)\,dt$

(c) After 20 years
\$11,353.35

35 \$85,750,000

39 (a) Less

(b) Can't tell

(c) Less

41 Future value more than \$20,000

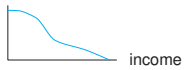
43 PV requires integral

47 5%, \$4761.90

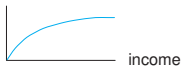
Section 8.7

1 (a)-(II), (b)-(I), (c)-(III)

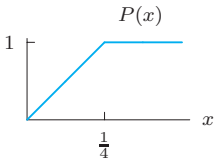
3 % of population per dollar of income



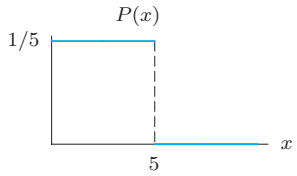
% of population having at least this income



5 pdf; 1/4



7 cdf; 1



9 cdf; 1/3

